DevSphere AI

UML and Workflow Handbook

Advanced Intelligent Software Engineering Assistant

Fourteen UML diagram families and the PostgreSQL ER model. Editable PlantUML, SVG and PNG files accompany this handbook.

Diagram index

01 Use case diagram

02 Class diagram

03 Object diagram

04 Component diagram

05 Composite structure diagram

06 Deployment diagram

07 Package diagram

08 Profile diagram

09 Activity diagram

10 State machine diagram

11 Sequence diagram

12 Communication diagram

13 Interaction overview diagram

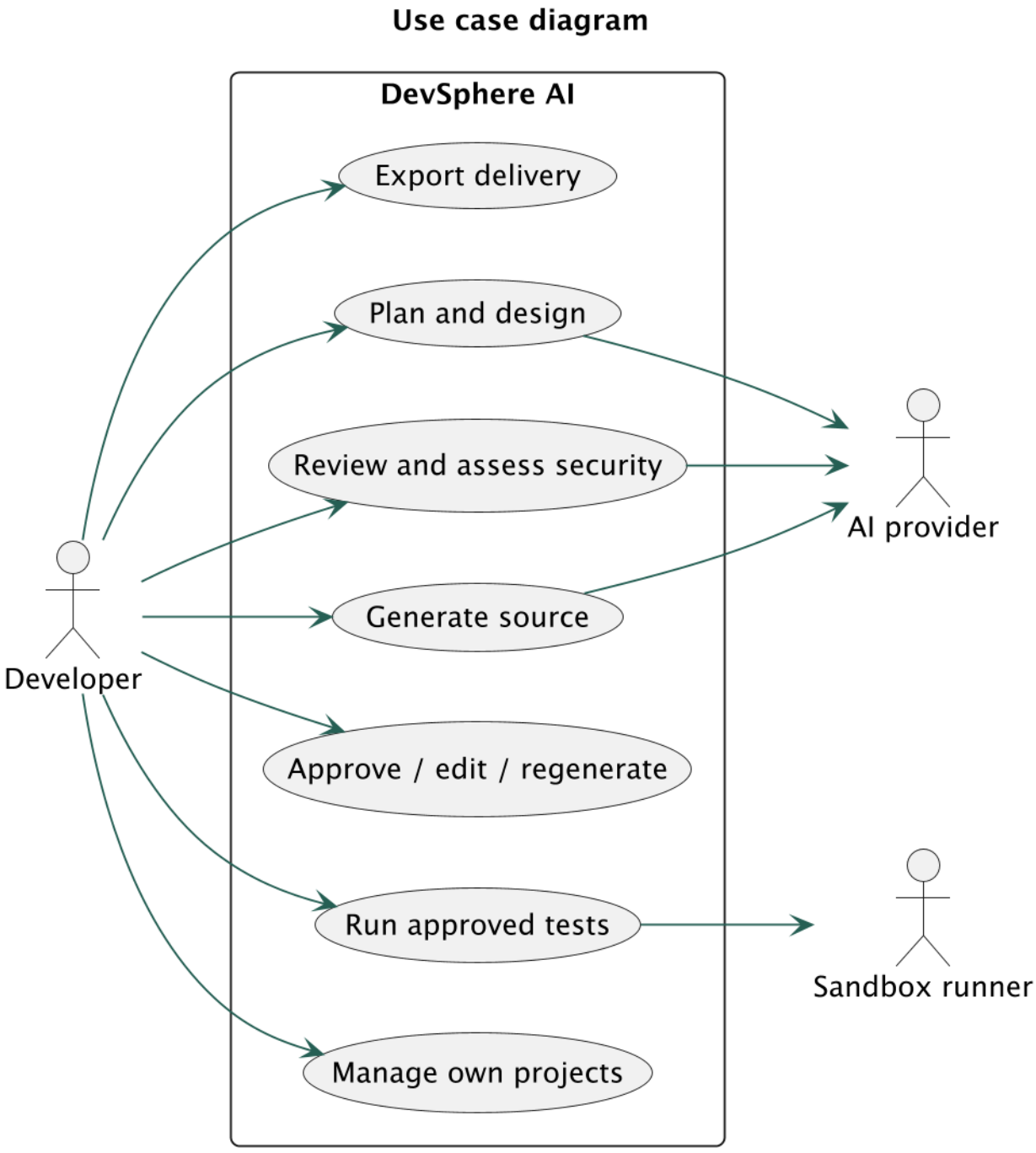
14 Timing diagram

15 Database ER diagram

Object and timing views are illustrative design examples. Actual runtime evidence is recorded separately in the testing report.

Use case diagram

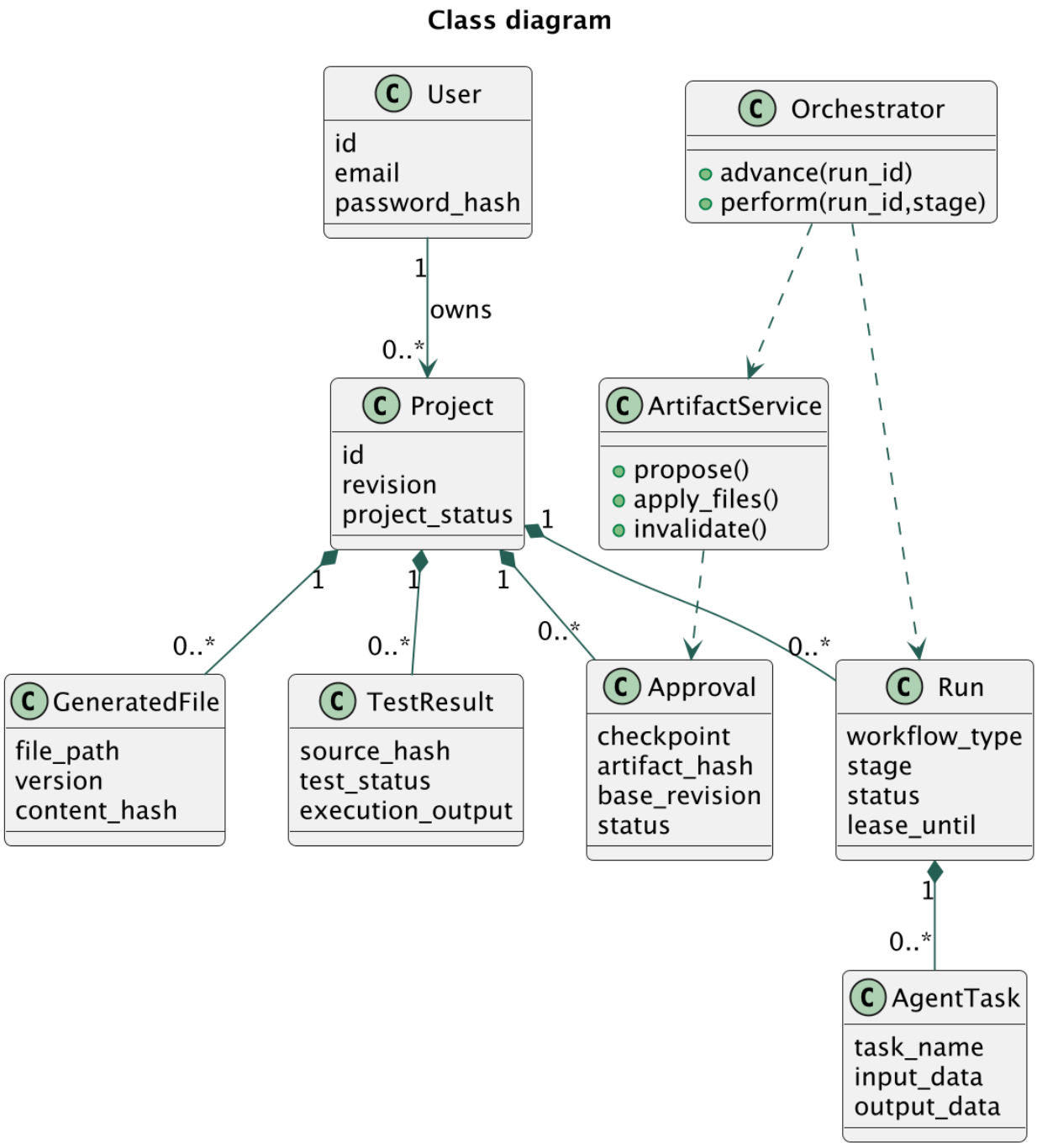
Shows developer actions and the boundaries of the AI provider and isolated runner.



Editable source: docs/diagrams/01-use-case.puml

Class diagram

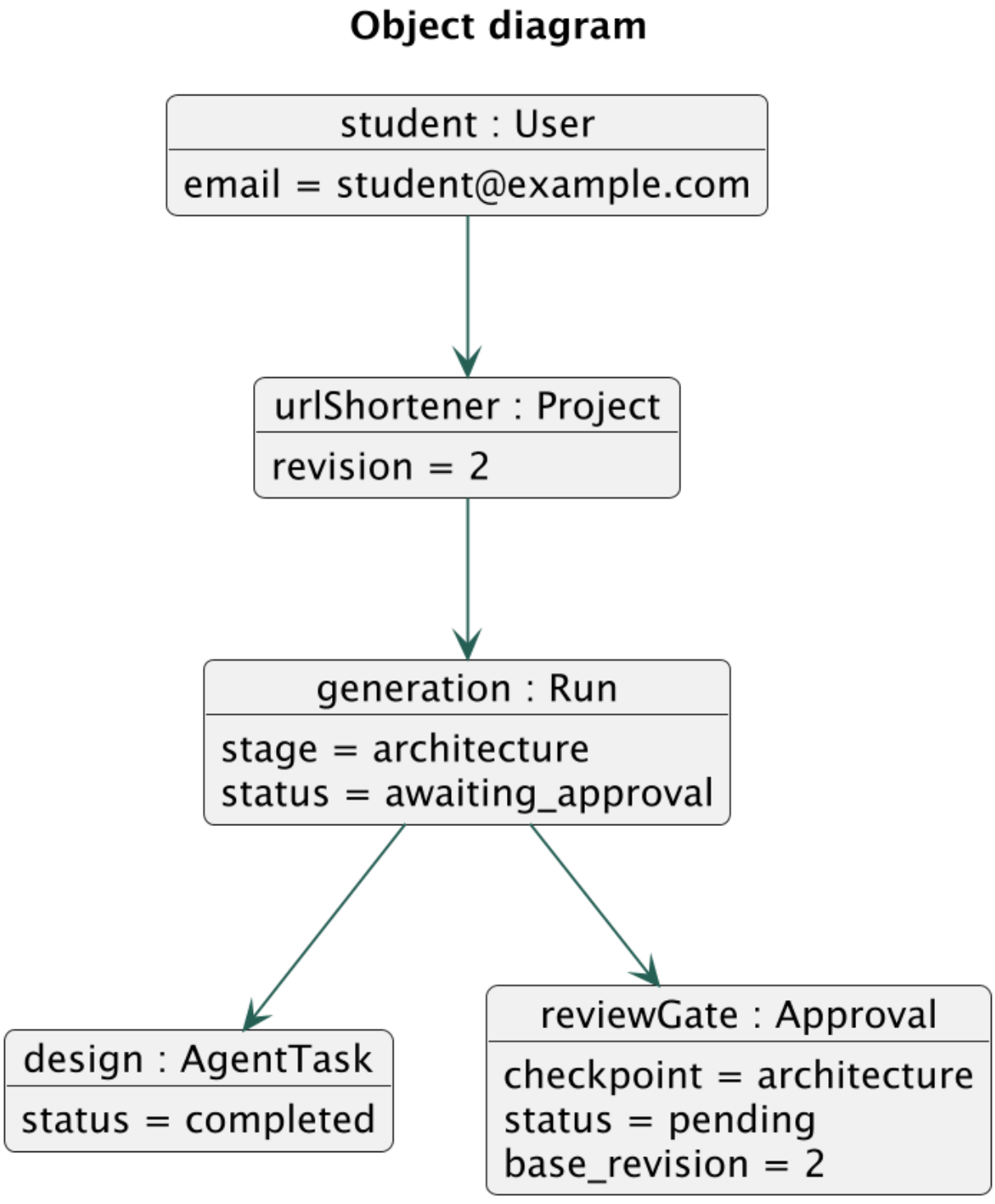
Core persisted entities and service responsibilities. The complete column dictionary is in the database schema.



Editable source: docs/diagrams/02-class.puml

Object diagram

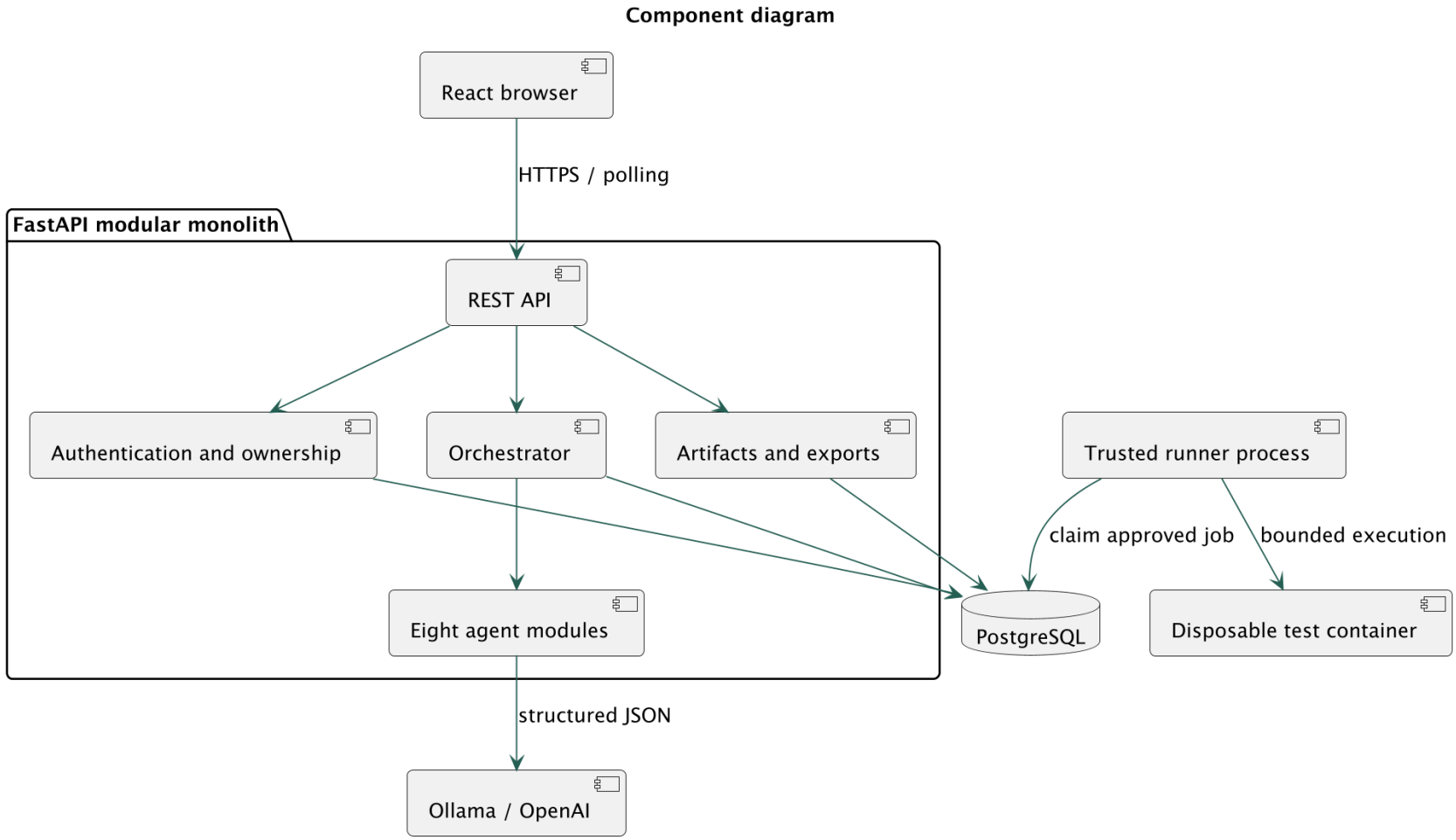
Illustrative snapshot of a workflow paused for architecture review; identifiers here are explanatory examples.



Editable source: docs/diagrams/03-object.puml

Component diagram

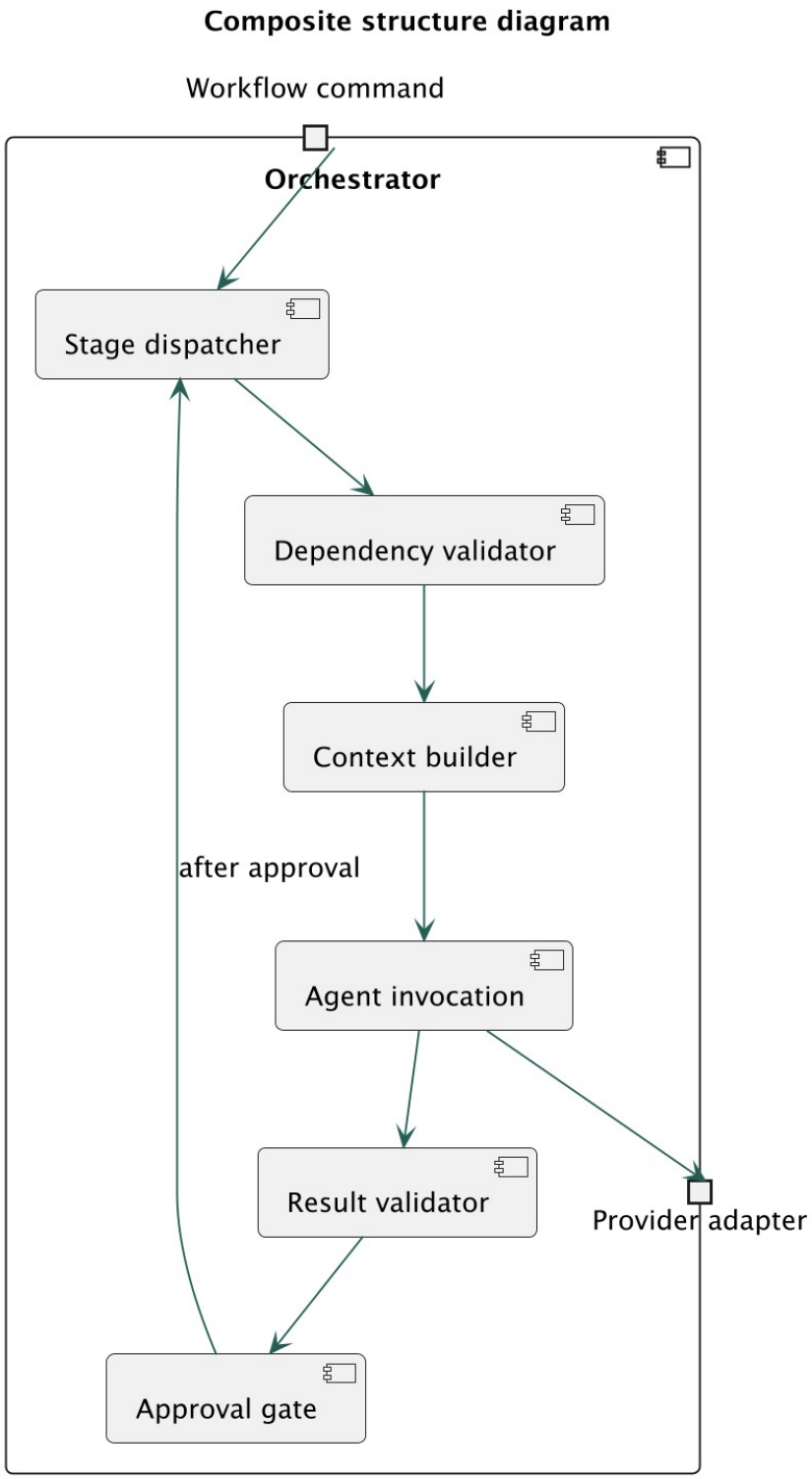
One modular application, one relational database and one separate trusted runner. Agent modules are Python functions.



Editable source: docs/diagrams/04-component.puml

Composite structure diagram

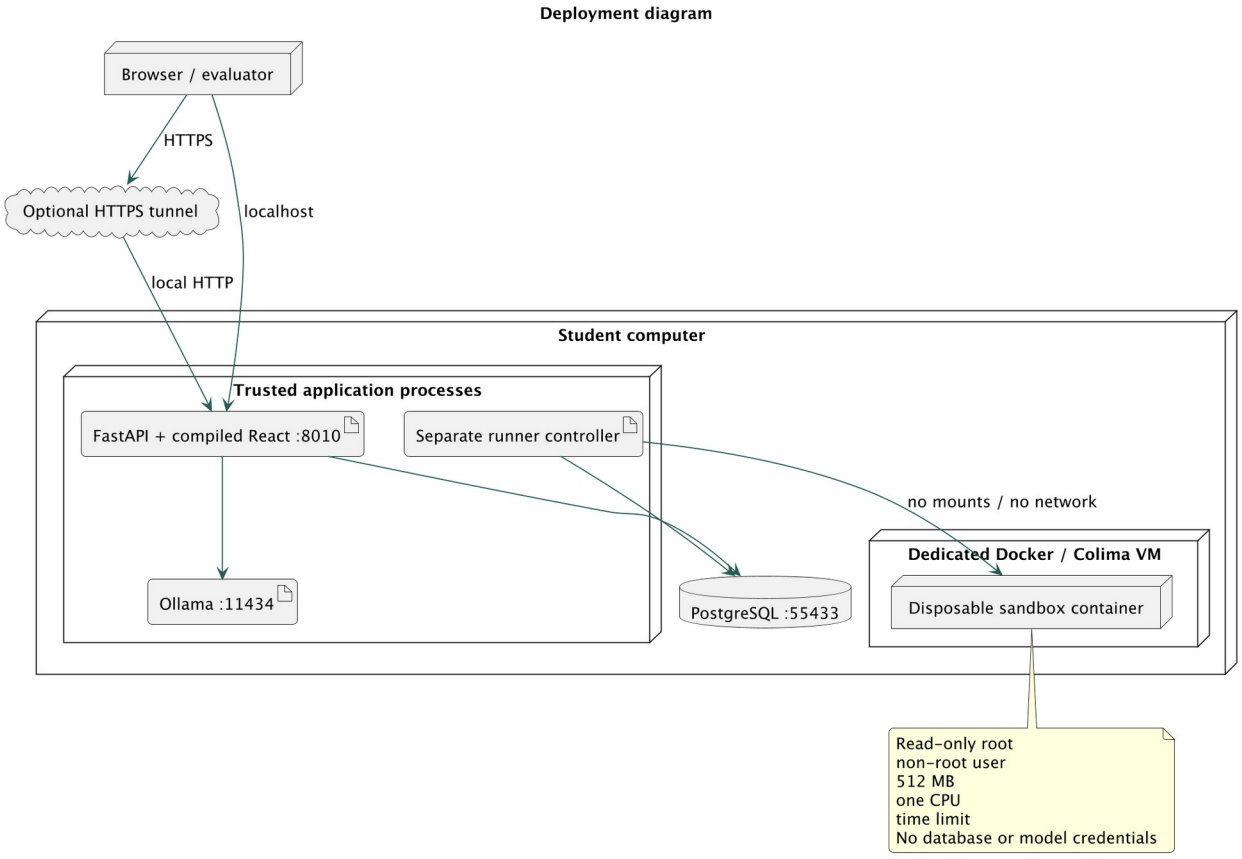
Internal collaboration of the orchestrator component; ports are logical interfaces rather than separate network services.



Editable source: docs/diagrams/05-composite-structure.puml

Deployment diagram

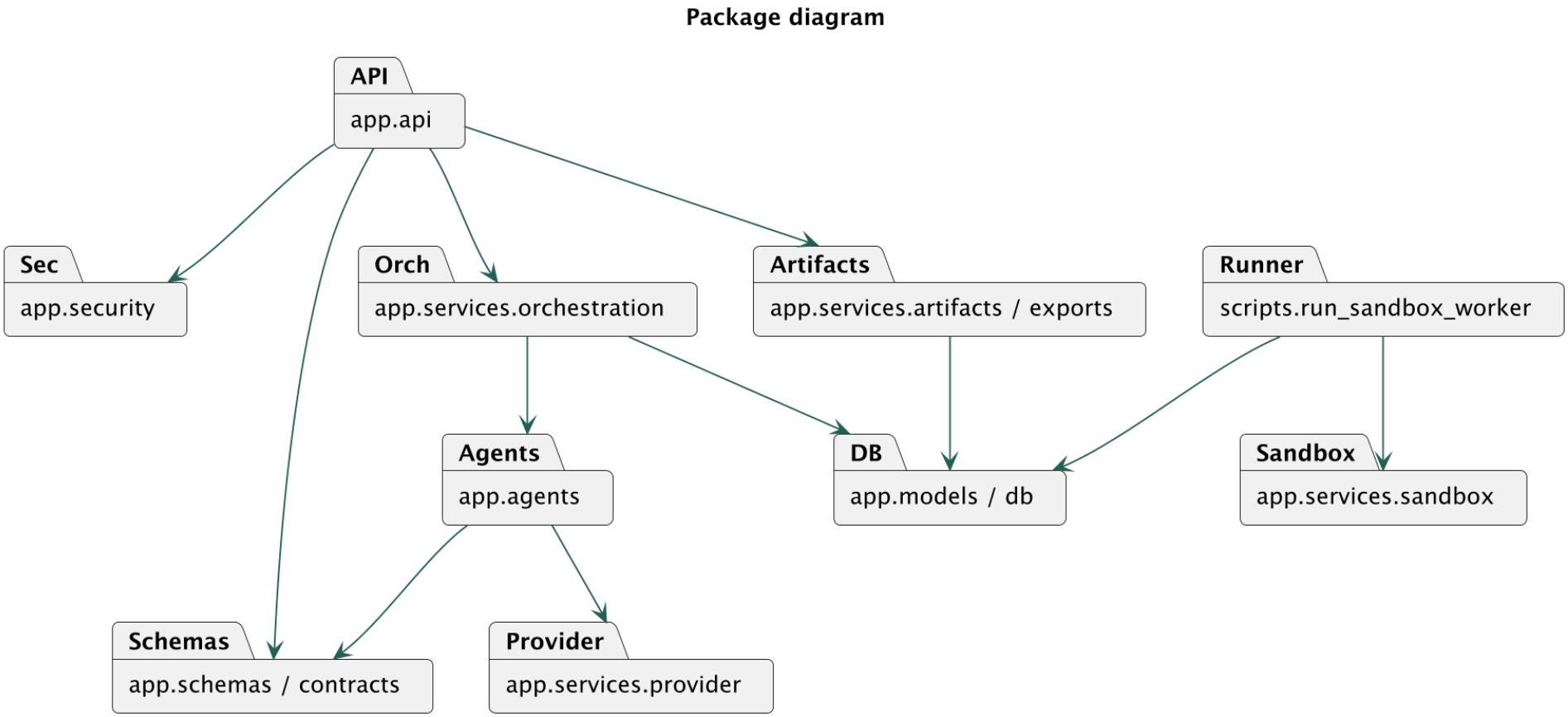
Local execution has a VM boundary for generated code. A free temporary tunnel exposes only the authenticated application.



Editable source: docs/diagrams/06-deployment.puml

Package diagram

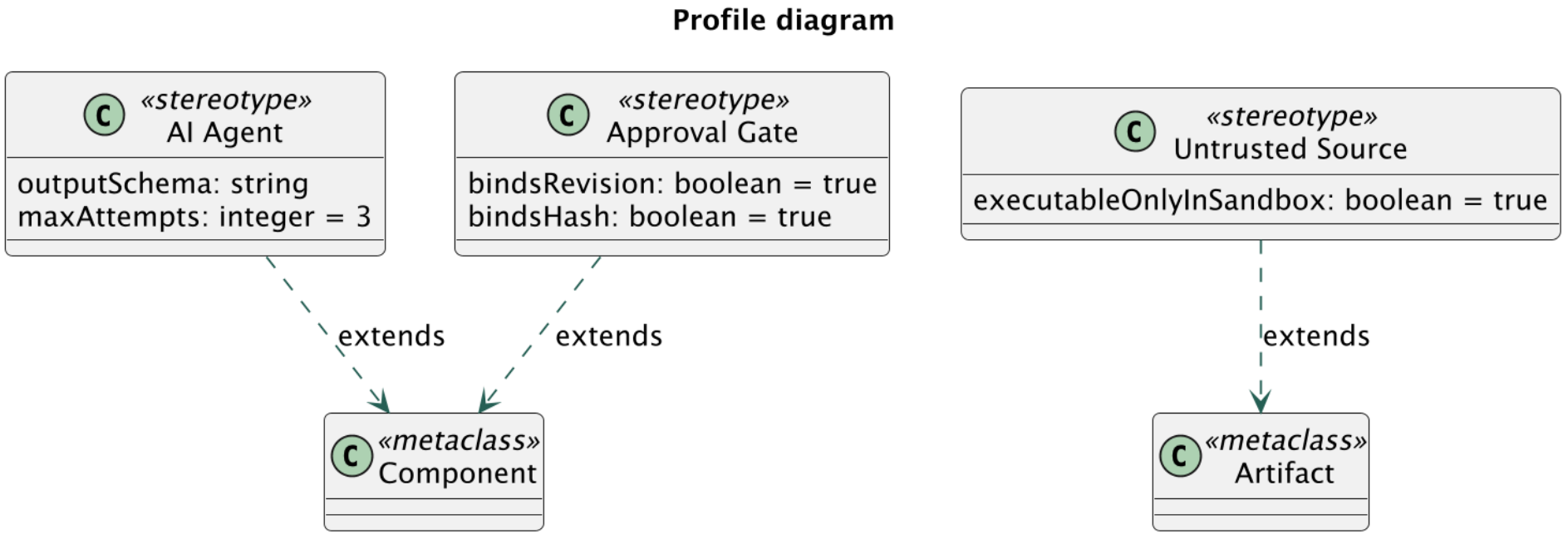
Source dependencies follow the modular monolith layout. The API never imports the sandbox executor.



Editable source: docs/diagrams/07-package.puml

Profile diagram

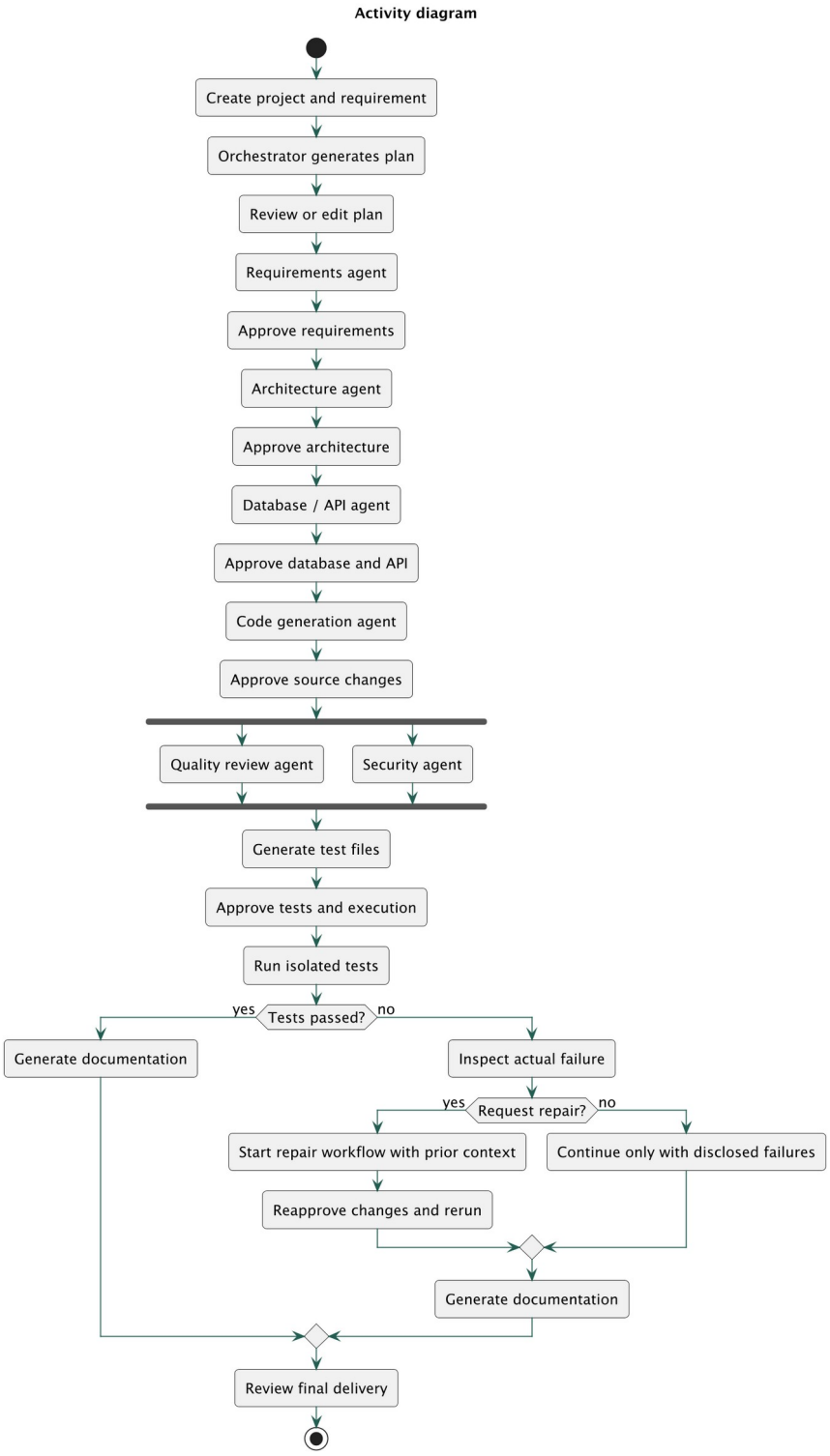
Design vocabulary used in this project. Stereotypes describe responsibilities; enforcement is implemented in validation, transactions and runner constraints.



Editable source: docs/diagrams/08-profile.puml

Activity diagram

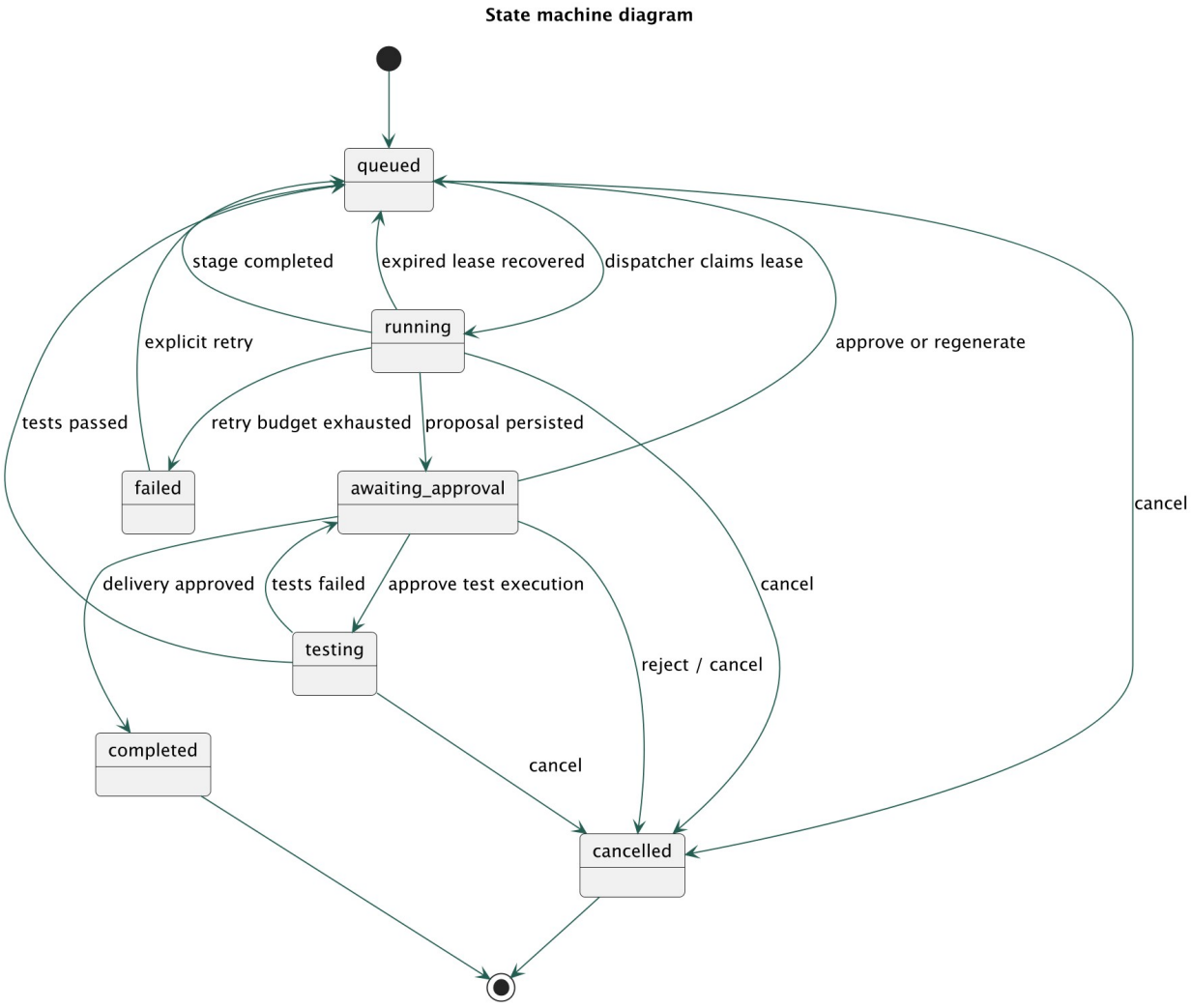
Full workflow includes explicit human gates and a conditional repair loop. Review-only workflows omit generation stages.



Editable source: docs/diagrams/09-activity.puml

State machine diagram

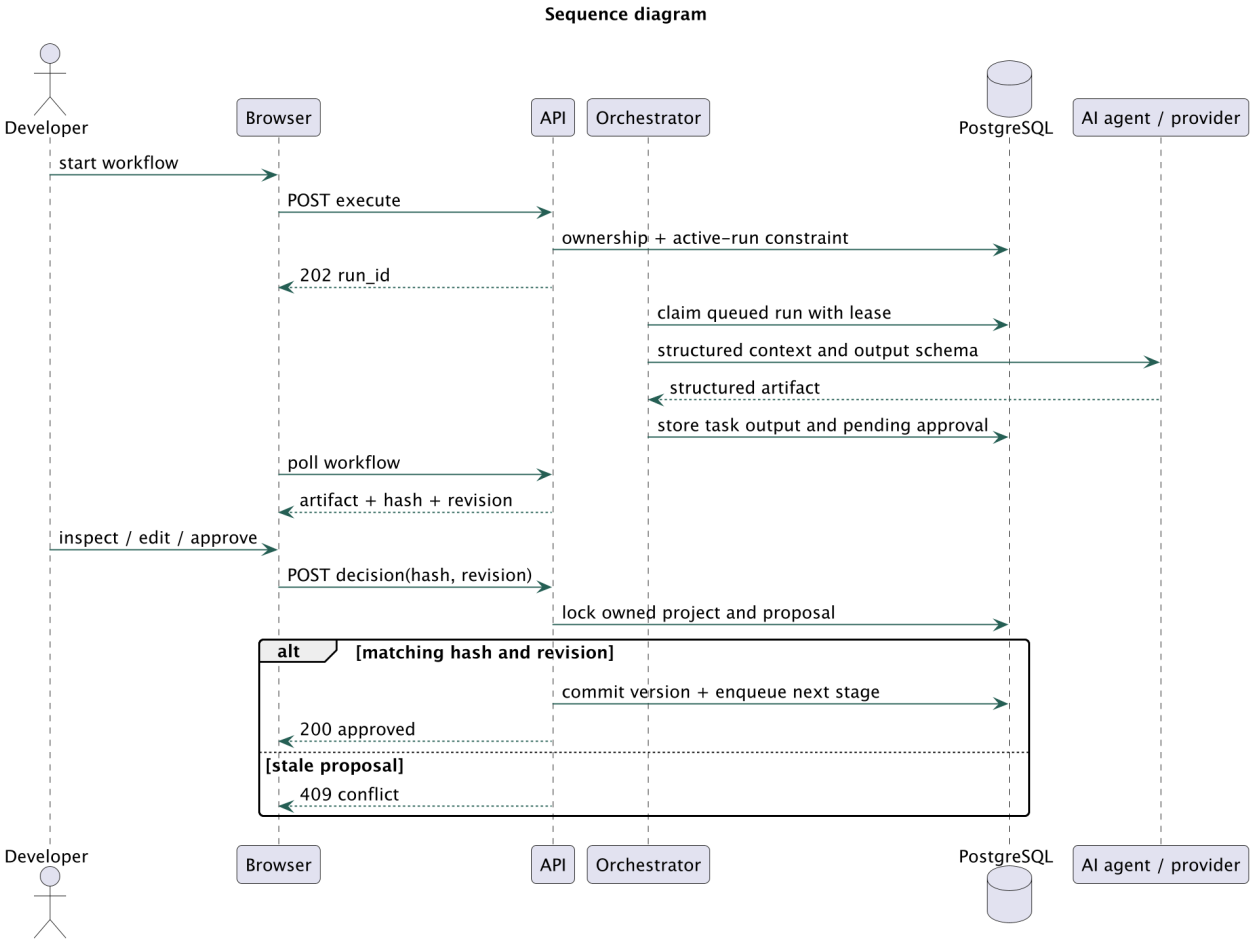
Persisted workflow status transitions, including a lease-expiry recovery path and cancellation.



Editable source: docs/diagrams/10-state-machine.puml

Sequence diagram

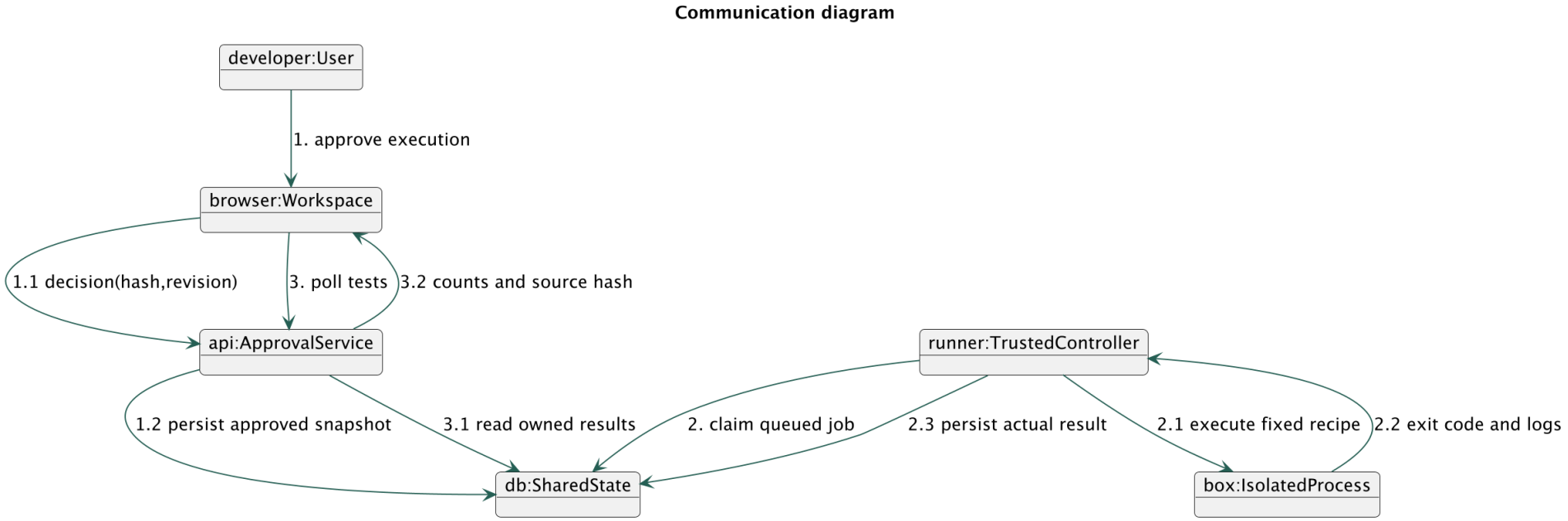
Optimistic approval binds both artifact content and project revision, preventing a stale browser action from committing different work.



Editable source: docs/diagrams/11-sequence.puml

Communication diagram

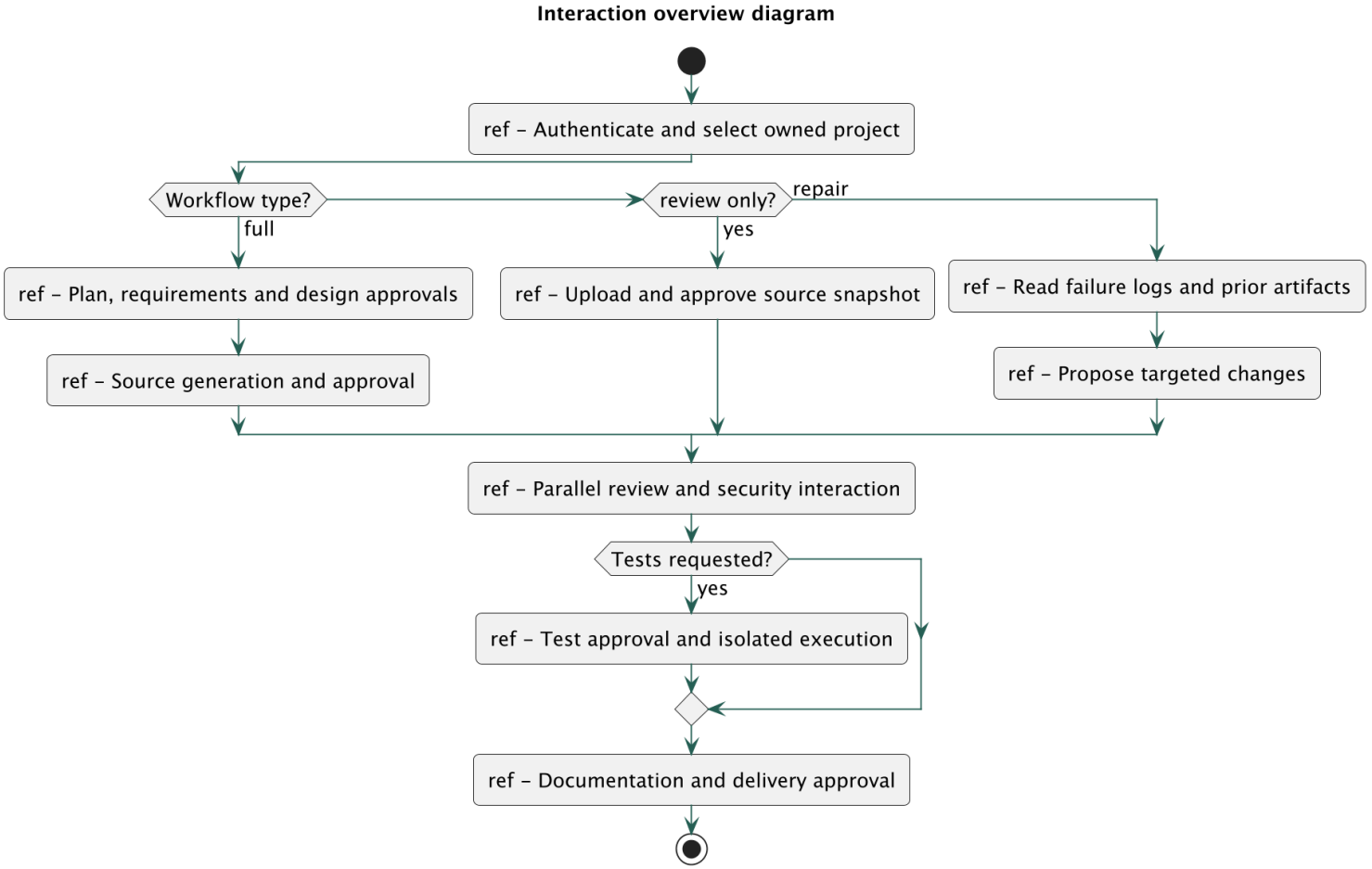
Numbered messages express collaboration around an approved test run. This uses object notation for UML communication links.



Editable source: docs/diagrams/12-communication.puml

Interaction overview diagram

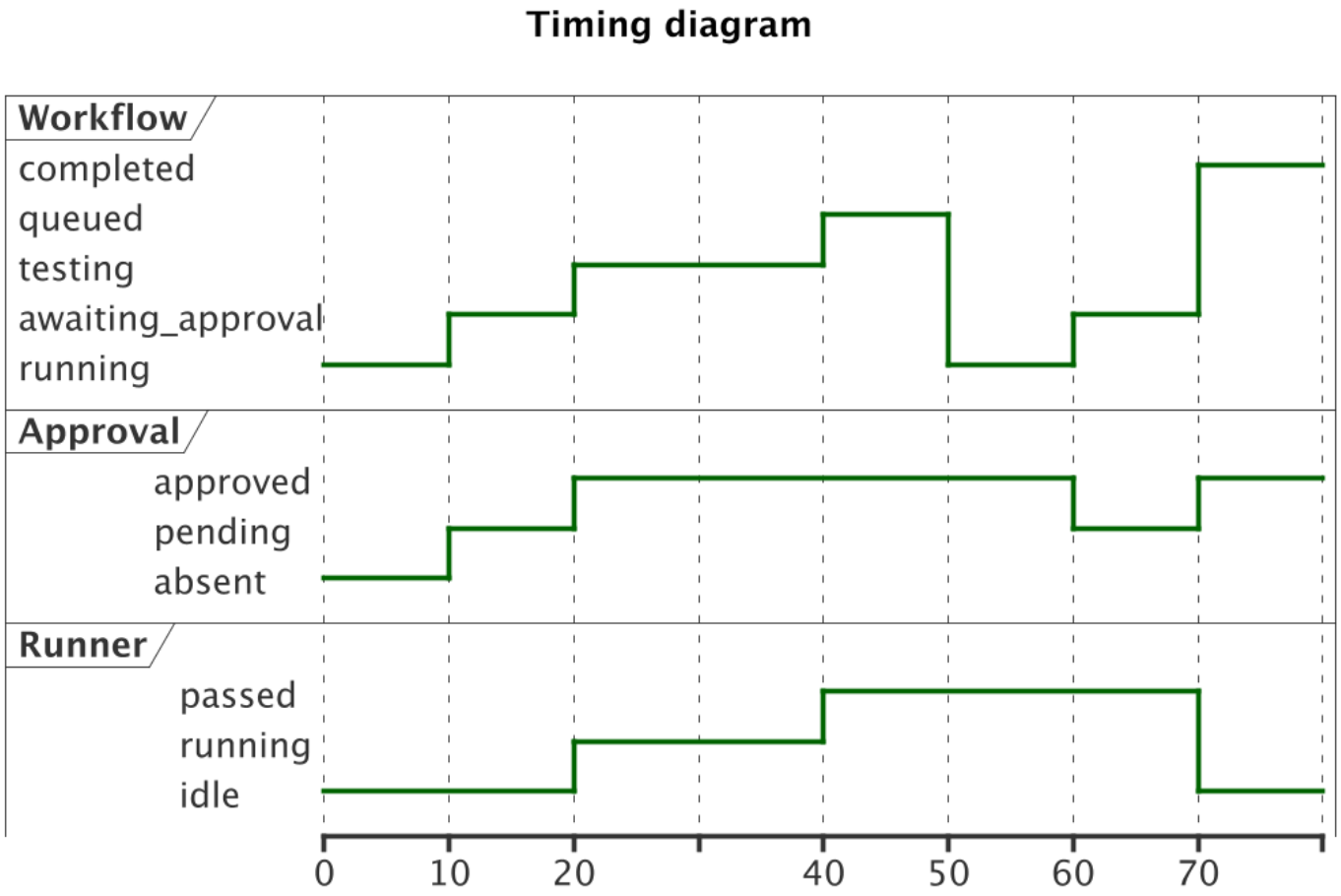
High-level references to detailed interactions. Decisions route complete generation, review-only or repair without running every agent.



Editable source: docs/diagrams/13-interaction-overview.puml

Timing diagram

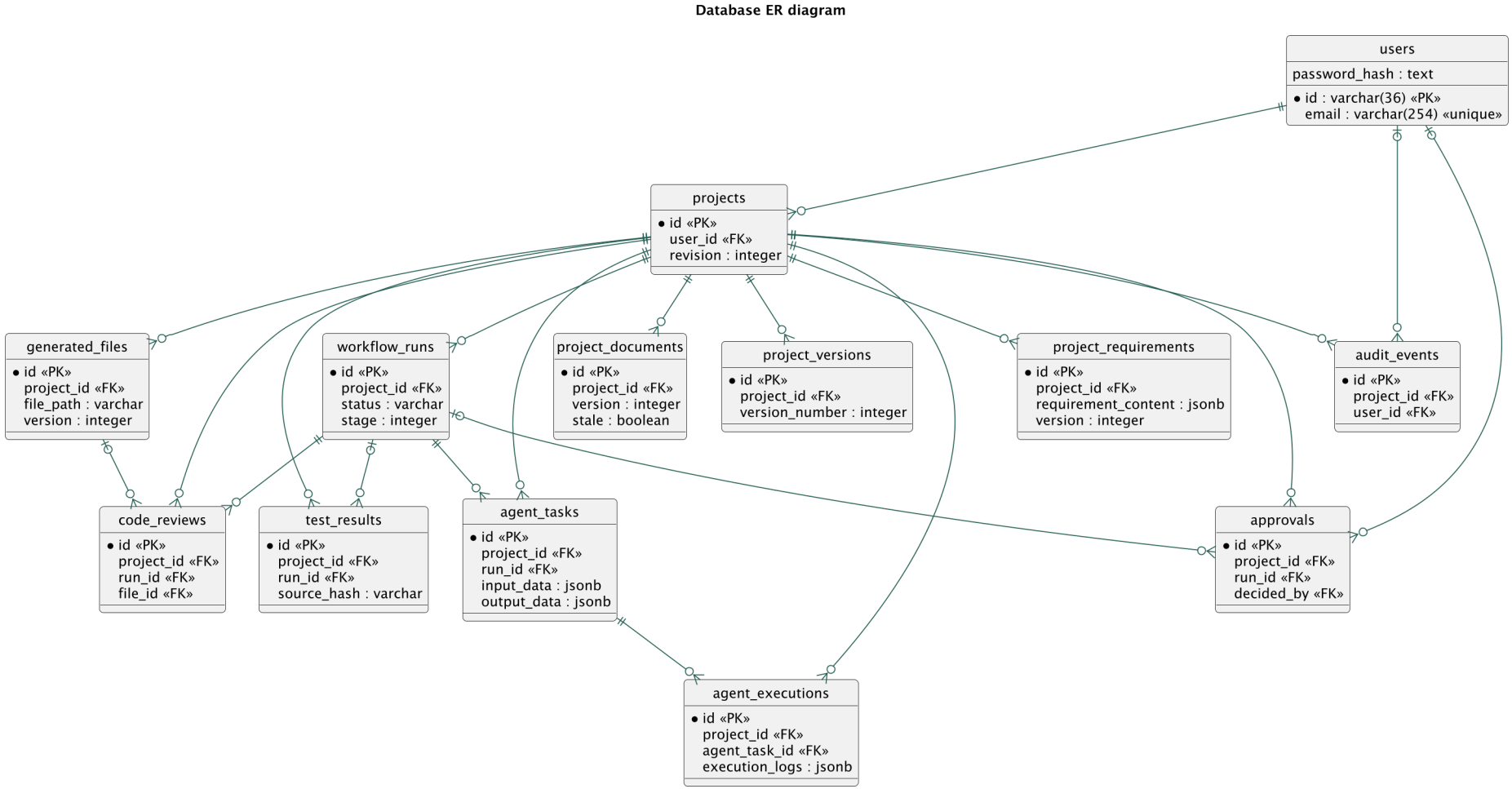
Illustrative ordering only; the horizontal units are logical ticks, not measured latency or a performance benchmark.



Editable source: docs/diagrams/14-timing.puml

Database ER diagram

All thirteen PostgreSQL tables are included. Project ownership is resolved through users and projects for every API operation.



Editable source: docs/diagrams/15-database-er.puml